2D Bottom-up micro-fabrication and characterization of a photodetector

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Abstract

The experience consists in the realization of two photodetectors, which produces variation in conduictivity when illuminated. The electrodes are deposited through photolitography process, in the interdigitated geometry with resolution of 1 μm. The semiconducting layer responsible for the production of new free carriers is $MAPbBr_3$, firstly spin coated and then crystallized directly from solution. The two photoconductors are realized by slow and fast crystallization. The I-V characteristics of both changes significantly when detector is irradiated by a light source.

Introduction

This experiment's purpose is to fabricate a photodetector device, composed by metal electrodes and a layer of polycrystalline perovskite. The electrode deposition is done via photolitography: a thin layer of photoresist is applied to the substrate by spin coating; then a template is written onto the photoresist via light pulses and the part that have interacted with light is washed away with particular solvents. The remaining photoresist is removed after metal evaporation and finally the perovskite layer is deposited. The photodetector consists in two electrodes separated by the semiconductor. The I-V characteristic is expected to be linear both in light and dark conditions, but the resistance depends on the density of free carriers. If the pohotodetector is illuminated with photons of energy higher than the energy gap, electron-hole pairs are generated and resistance is decreased. The photodetector then behaves as a photoconductor. The electrode's geometry is interdigitated in order to increase the gain: Fig. (1) shows that the distance betweed electrodes is reduced, conserving the area where carriers can be generated.

Materials and design

The perovskite crystal used to fabricate the photodetector is methylammonium lead bromide ($MAPbBr_3$), an hybrid semiconductor with band gap of 2.2 eV, which corresponds to the yellow-green light. The substrate is a 25×25 mm^2 microscope glass. The procedure consists in the following steps:

- **Substrate preparation**: impurities are removed from the glass with solvent bath, followed by ultrasonic bath for 10 minutes. The last ultrasonic bath is done with distilled water, then the substrate is blow dried with nitrogen. The adsorbed water is removed placing the glass onto an hot plate at 110 °C for 5 minutes.
- **Photoresist Spin-coating** [**1**]: 450 μL of S1818 photoresist is placed on the cleaned surface of the glass with a micropipette. The spin

coater is set at 4000 rpm, 1 s ramp, 60 s dwell. After finishing, the sample is placed on the hot plate at 110 °C for 1 minute. From this step is necessary to work with yellow light condition, because the photoresist is sensible to blue light

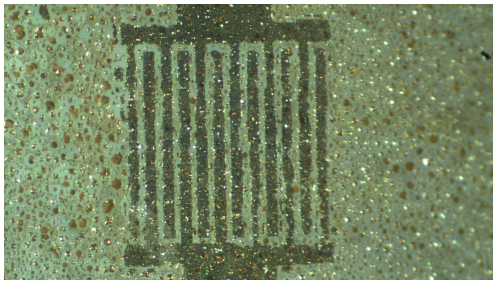
- **Photoresist writing** [**2**]: The exposition is done with 1 μm resolution for the electrodes and 5 μm for the contact pads. The pattern emerges after two subsequent bath of MF-319 developer and distilled water.
- **Metal evaporation**: Au has high conductivity and is resistant against air oxidation, but it is difficult to attach it to the glass substrate. For this reason also Cr is used. The evaporator works in UHV condition.
- **Lift-Off**: placing the sample in acetone for 4 hours, the remaining photoresist is removed. The sample is plasma cleaned in order to ensure a good perovskite deposition.
- **Perovskite deposition**: Two photodetectors are realized with different techniques. 100 μL of perovskite solution are distributed onto the substrate in both cases. In the former, the sample is simply spin coated for 30 s at 800 rpm. With this method the crystallization occurs slowly. The latter is called antisolvent technique, in which the spin coater works at 800 rpm for 15 s and Chlorobenzene is added during a 2000 rpm coating which lasts for another 15 s. In this case the crystallization occurs faster. The two samples are finally placed on the hot plate at 100 °C for 30 minutes.

Results and Discussion

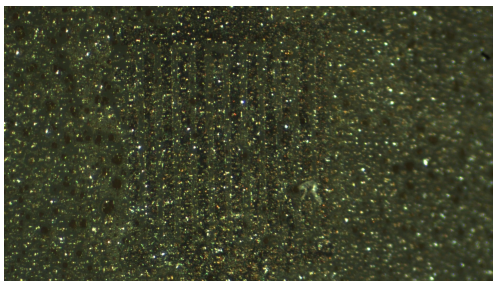
The two samples can be inspected simply with an optical microscope. The sample obtained with antisolvent technique appears more opaque: crystal are smaller and less spaced. This is because more time is required in order to grow a larger crystal. Fig. (1a) and Fig. (1c) are taken with the same magnification, it is evident that in the first figure crystal are larger. The devices performances are measured analysing the I-V characteristics. Inside the probe station, electrical contact are made and the device is biased from -10 V to 10 V and form 10 V to -10 V. We expect a variation in the characteristics when light is turned on. From Fig. (2) it's clear that the conductivity increases if light is turned on. The non-linear and hysteretic behaviour is due to imperfections during the preparation steps.

Conclusions

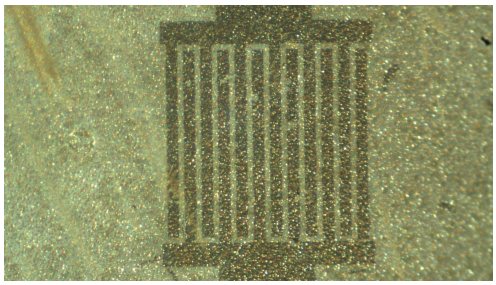
Fig. (2) demonstrates that conductivity of theese detectors changes in light conditions, but a lot of noise is present. From Fig. (3) the effect of light switching is more evident: the current increase by approximately two orders of magnitude.



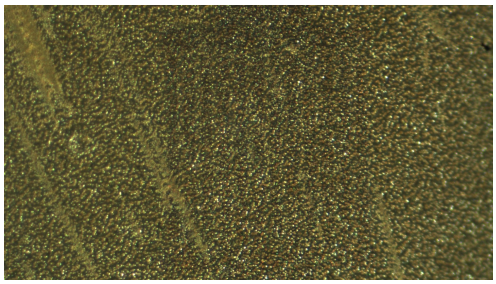
(a)



(b)

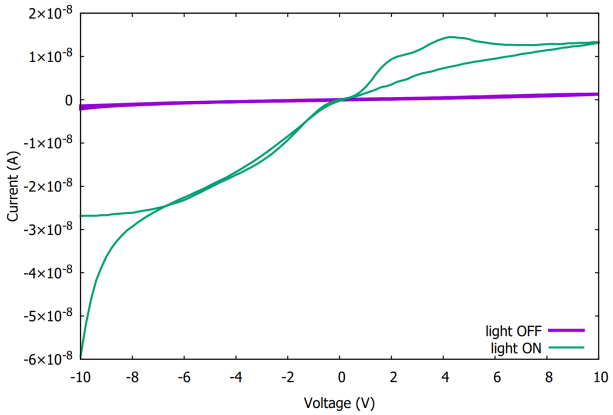


(c)

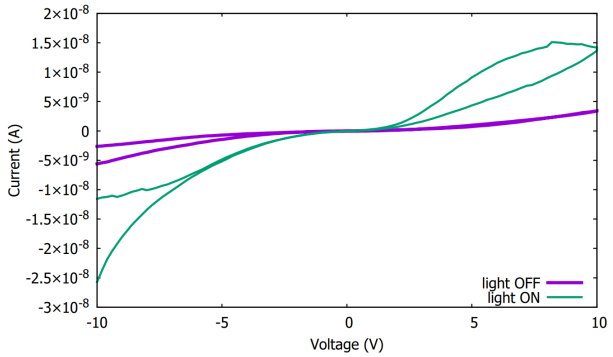


(d)

Figure 1: (a)-(b) are taken from the normal photodetector; (c)-(d) are from the antisolvent one. Pictures are taken under light and dark condition



(a)



(b)

Figure 2: I-V characteristics of (a) normal photodetector and (b) antisolvent one.

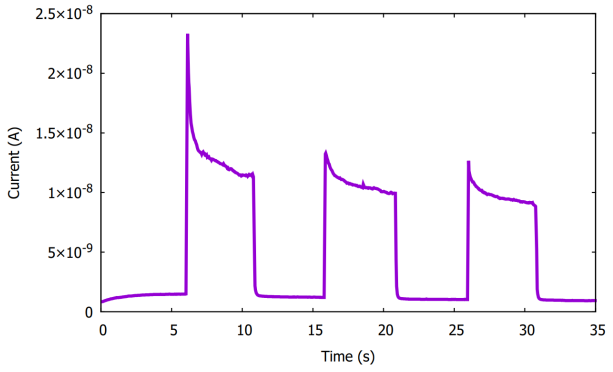


Figure 3: Current at 10 V. The variation of current between 10^{-10} to 10^{-8} A is due to light switching.

References

[1] *6800 spin coater series*. URL: https://scscoatings.com/equipment/%20spin-coating-systems/6800-spin-coaters/.

[2] *MicroWriter ML3 Pro*. URL: https://www.durhammagnetooptics.com/%20?product=microwriter.